

ELITE FULL STACK ENGINEERING ROADMAP · 2025-2026

First-Year CSIT Student · Full Stack + Backend Engineering · 12-Month Internship Sprint

CSIT Yr 1

Full Stack Target

12 Months

Internship Ready

 CURRENT POSITION

Strong Foundation. Ready to Build.

Final trimester of Yr 1 CSIT. Java fundamentals solid. DSA through Trees completed. HTML/CSS/JS + Git in place. Currently learning React — ahead of most peers at this stage.

Java ✓

HTML/CSS ✓

JS ✓

DSA Trees ✓

React (in progress)

 1-YEAR VISION

Internship-Ready Full Stack Dev

Ship a full-stack SaaS project. Master React + Node.js + PostgreSQL. Complete 300+ DSA problems. Build a GitHub profile that speaks before you do. Land a Software Engineering internship.

MERN Stack

REST APIs

Auth Systems

Deployment

 STRATEGIC EDGE

Your Unfair Advantages

OOP via Java = backend mental model already built. Web trifecta (HTML/CSS/JS) means frontend is not starting from zero. DSA progress = you're ahead of 70% of CS freshers.

OOP Mindset

Early Starter

DSA Head Start

Skill Area	Current Level	Required Level	Gap / Notes	Priority
React.js	●●○○○	●●●●○	Actively learning — maintain momentum	HIGH
Advanced JavaScript	●●○○○	●●●●○	Async/await, closures, ES6+ needed	HIGH
Node.js + Express.js	○○○○○	●●●●○	Full backend stack — critical gap to close	HIGH
REST APIs	○○○○○	●●●●○	Design + consume APIs end-to-end	HIGH
Databases (SQL + NoSQL)	○○○○○	●●●●○	PostgreSQL + MongoDB fundamentals	HIGH
Authentication + Security	○○○○○	●●●●○	JWT, OAuth, bcrypt — interview essential	MED
DSA (Beyond Trees)	●●●○○	●●●●●	Graphs, DP, Heap, Backtracking	HIGH
Deployment + DevOps Basics	○○○○○	●●●●○	Vercel, Render, Railway + basic CI/CD	MED
System Design Basics	○○○○○	●●○○○	HLD intro for internship interviews	LOW
Git / GitHub Advanced	●●○○○	●●●●○	Branching, PRs, open source workflow	MED

PHASE 1

JS Deep Dive

ES6+, async/await, closures, DOM mastery. Finish React: hooks, state, routing, API calls

PHASE 2

Backend Core

Node.js + Express. REST APIs, MVC pattern. Middleware, CRUD, error handling

PHASE 3

Databases

PostgreSQL + Prisma ORM. MongoDB + Mongoose. Schema design, indexing, queries

PHASE 4

Full Stack

Auth (JWT/OAuth). Full MERN applications. State management. API integration

PHASE 5

Deploy + DSA

CI/CD, Vercel/Render. Graphs, DP, Heap. 300+ LeetCode. System design basics

OUTCOME

Internship ✓

Portfolio live, GitHub active, LinkedIn strong, offers in hand

 PORTFOLIO PROJECTS — PROGRESSIVELY DIFFICULT

PROJECT 01 · Beginner

Personal Dev Portfolio

React

CSS Modules

Framer Motion

Vercel

First live URL + online presence

PROJECT 02 · Easy+

Task Management App

React

Node.js

Express

MongoDB

First full CRUD + REST API

PROJECT 03 · Medium

Auth System (JWT)

JWTbcryptPostgreSQLPrisma

✦ Security skills = high resume value

PROJECT 04 · Medium+

Full Stack Blog Platform

MERNRich TextAuthImage Upload

✦ Full product thinking on display

PROJECT 05 · Industry-Level

Real-Time SaaS Dashboard

WebSocketsRedisDockerCI/CD

✦ Standout — interview centrepiece

🌐 NETWORKING & PERSONAL BRANDING BLUEPRINT

LinkedIn Strategy

► Post 2-3x/week: project updates, learnings, code snippets

► Connect with 10 engineers/week from target companies

► Write 1 long-form post/month about a project you built

► Engage on job posts with thoughtful comments

► Target: 1,000+ connections by Month 9

GitHub Strategy

► Commit daily — keep the green squares alive

► Pin 3 best projects with screenshots + live demos

► Write detailed READMEs for every project

► Contribute to 1 open source project by Month 7

► Star + fork relevant repos to signal interest

Community + Content

► Join: 100xDevs, The Odin Project Discord, JS India

► Attend 2 hackathons (online or offline)

► Post dev threads (build in public on LinkedIn/Twitter)

► Document your learning journey weekly

► Find 1 accountability partner or study group

📅 MONTHLY MILESTONES

Month	Focus Area	Deliverable	Outcome
Month 1	Finish React + JS Deep Dive	Portfolio website live (Project #1)	First deployed app, GitHub properly set up
Month 2	Node.js + Express fundamentals	REST API (CRUD) built from scratch	Backend thinking unlocked, first server running
Month 3	MongoDB + Task Manager App	Project #2 live with full CRUD	First full-stack app, MERN foundation complete
Month 4	PostgreSQL + Auth system	Project #3: JWT Auth API + frontend	Can build secure authenticated applications
Month 5	Full Stack Blog Platform	Project #4 live with all features	Portfolio now shows full product development
Month 6	DSA Sprint: Graphs + DP + Heap	100+ LeetCode (200+ total)	Interview-level DSA coverage expanding
Month 7	Open Source + Deployment	First OSS PR merged, all projects deployed	GitHub looks active, professional, credible
Month 8	SaaS Dashboard (Industry Project)	Project #5 started, 60% complete	Building most advanced portfolio project
Month 9	Project #5 complete + Applications	SaaS project live, 20 applications sent	Portfolio complete, applications actively out
Month 10	Mock Interview + DSA Sprint	300+ LeetCode, 15 mock interviews done	Interview confidence high, technical speed improving
Month 11	Full Interview Mode	50+ applications, referrals pursued	Interview calls coming in, feedback loop active
Month 12	Negotiate + Accept Offer	Internship / Job offer signed	🏆 Goal achieved — engineering career launched

🔥 IMMEDIATE ACTION PLAN

⚡ Next 24 Hours

•Restructure GitHub — add bio, photo, pinned repos

•Optimize LinkedIn: add "Open to Internships" banner

•Solve 3 LeetCode Easy problems (build the habit)

•List existing projects — push all to GitHub

•Start Portfolio website design or direct code

 Next 7 Days

- Complete React: useEffect, custom hooks, React Router
- Study JS: Promises, async/await, event loop, closures
- Deploy portfolio v1 to Vercel
- Solve 15 LeetCode (10 Easy + 5 Medium)
- Connect with 15 developers on LinkedIn

 Next 30 Days

- Portfolio website fully live with projects section
- Start Node.js: set up Express server + 3 endpoints
- Solve 60+ LeetCode problems total
- Post 4 LinkedIn updates (1/week — learning + project)
- 100+ GitHub commits in the streak



SUCCESS METRICS AT 12 MONTHS

5

Live Portfolio Projects

300+

LeetCode Solved

365+

GitHub Commits

1K+

LinkedIn Connections

50+

LinkedIn Posts

1

Internship Offer 